Def. Let V be a vector space,
Given a vector $\alpha \in V$ and linear operator $T : V \to V$, the subspace

$$Z(\alpha, T) \overset{\text{def}}{=} \{ g(T)\alpha \mid g(t) \in F[t] \} = \operatorname{span}\{ T^k \alpha \mid k \in \mathbb{N} \cup \{0\} \}.$$

is called the T-cyclic subspace generated by α

Def. Let V be a vector space. Given $\alpha \in V$ and a linear operator T on V, an ideal

$$M(\alpha, T) = \{ g(t) \in F[t] \mid g(T)\alpha = 0 \}$$

is called the T-annihilator of α, and the unique monic generator of also called ".

Thm. Let α be any non-zero vector in V and let P_α be the T-annihilator of α.
1) $\deg P_\alpha = \dim Z(\alpha, T)$
2) $Z(\alpha, T) = \operatorname{span}\{ T^k \alpha \mid 0 \leq k \leq \deg P_\alpha \}$
3) The minimal polynomial of $T|_{Z(\alpha,T)}$ is P_α.
Proof. Given a polynomial $g \in F[x]$, the Euclidean Algorithm gives

$$g = P_\alpha \cdot q + r \quad \text{for some } q, r \in F[x], \text{ with } r = 0 \text{ or } \deg r < \deg P_\alpha.$$

Then, $g(T)\alpha = P_\alpha(T) q(T)\alpha + r(T)\alpha = r(T)\alpha$. Since either $r=0$ or $\deg r < \deg P_\alpha$, the vector $r(T)\alpha$ is a linear combination of $\{\alpha, T\alpha, T^2\alpha, \cdots, T^{\deg P_\alpha - 1}\alpha\}$.
Since g was arbitrary, $Z(\alpha, T)$ is spanned by $\{ T^n \alpha \mid n = 0, 1, \cdots, \deg P_\alpha - 1 \}$
Moreover, these are linearly independent, because; if there exists non-trivial representation

$$c_0 \alpha + c_1 T\alpha + \cdots + c_{\deg P_\alpha - 1} T^{\deg P_\alpha - 1} \alpha = 0,$$

then it contradicts to the minimality of P_α. It proves 1) and 2),
To show the third assertion, let $U = T|_{Z(\alpha,T)} : Z(\alpha, T) \to Z(\alpha, T) : \alpha \mapsto T\alpha$.
It is well-defined, since the cyclic subspace is invariant under T.
Given a polynomial $g \in F[x]$,

$$P_\alpha(U) g(T)\alpha \underset{\substack{\uparrow \\ g(T)\alpha \in Z(\alpha, T)}}{=} P_\alpha(T) g(T)\alpha = g(T) P_\alpha(T)\alpha = g(T)0 = 0.$$

So, $P_\alpha(U)$ is identically zero. Moreover, $h(U)$ cannot be zero if $\deg h < \deg P_\alpha$, because $h(U)\alpha = h(T)\alpha = 0$, then it contradicts to the minimality.

「Companion Matrix.

Let W be a vector space of dimension $k \in \mathbb{N}$, and $T: W \to W$ be a linear operator which have a cyclic vector $\alpha \in W$.

By the Theorem above, $\{\alpha, T\alpha, \cdots, T^{k-1}\alpha\}$ forms a basis for W.

As above fact and the Cayley-Hamilton Theorem, let

$$P_\alpha(x) = x^k + c_{k-1}x^{k-1} + \cdots + c_1 x + c_0$$

be the minimal polynomial (and also characteristic) for T.

Define $\beta = \{T^{n-1}\alpha \mid n = 1, 2, \cdots, k\}$, denote $\alpha_n = T^{n-1}\alpha$. Then,

$$0 = 0 \cdot \alpha = P_\alpha(T)\alpha = T^k\alpha + c_{k-1}T^{k-1}\alpha + \cdots + c_1 T\alpha + c_0\alpha$$

$$\Rightarrow T^k\alpha = -c_0\alpha - c_1 T\alpha - \cdots - c_{k-1}T^{k-1}\alpha$$

$$= -c_0\alpha_1 - c_1\alpha_2 - \cdots - c_{k-1}\alpha_k$$

$$[T]_\beta = \begin{bmatrix} 0 & 0 & \cdots & 0 & -c_0 \\ 1 & 0 & \cdots & 0 & -c_1 \\ 0 & 1 & \cdots & 0 & -c_2 \\ \vdots & & \ddots & \vdots & \vdots \\ 0 & \cdots & 0 & 1 & -c_{k-1} \end{bmatrix} = \begin{bmatrix} 0 & \cdots & 0 & -c_0 \\ & I_{k-1} & & \vdots \\ & & & -c_{k-1} \end{bmatrix}.$$

Directly, we obtain that: on a fin. dim space T,

A linear operator T has a cyclic vector

if and only if

the minimal polynomial of T and characteristic polynomial of T are identical.

Exercises in Hoffman, Sec 7.2

1. Every linear operator T on two-dimensional either;

T has a cyclic vector or T is a scalar multiple of the identity.

Proof. Suppose that T has no cyclic vector. That is, for all $\alpha \in V$, $\{\alpha, T\alpha\}$ is linearly dependent. Therefore, there is $c \in F$ such that $c\alpha = T\alpha$. Let $\alpha_1 \in V$ be a non-zero vector. For some $c_1 \in F$, $c_1\alpha_1 = T\alpha_1$.
Now, let $\alpha_2 \in V \setminus \langle \alpha_1 \rangle$, Then, similarly, $c_2\alpha_2 = T\alpha_2$, for some $c_2 \in F$.
By the construction, $\alpha_1 - \alpha_2 \neq 0$, and there is $c_3 \in F$ such that $c_3(\alpha_1 - \alpha_2) = T(\alpha_1 - \alpha_2) = c_1\alpha_1 - c_2\alpha_2$, so

$$(c_3 - c_1)\alpha_1 = (c_3 - c_2)\alpha_2$$

But, since $\alpha_2 \in V \setminus \langle \alpha_1 \rangle$, $c_3 - c_1 = c_3 - c_2 = 0$, that is, $c_1 = c_3 = c_2$.
Now, since $\{\alpha_1, \alpha_2\}$ is basis for V, $T = c_1 I$. ☒

4. If T^2 has a cyclic vector, then T has a cyclic vector.
Proof. Let $\alpha \in V$ be the cyclic vector of T^2. That is, $\mathcal{B} = \{T^{2k}\alpha \mid k = 0, 1, \cdots\}$ spans the space V. Clearly, $\mathcal{B}' = \{T^k\alpha \mid k = 0, 1, \cdots\}$ satisfies $\mathcal{B} \subseteq \mathcal{B}'$, therefore $V = \langle \mathcal{B} \rangle \subseteq \langle \mathcal{B}' \rangle = V$.

But the converse does not hold; let $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $\alpha = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Then,
$\{\alpha, A\alpha\} = \left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}\right\}$ spans F^2 but

$A^2 \equiv 0$ has no cyclic vector.

5. Let V be an n-dimensional over F, and let N be a nilpotent on V. Suppose that $N^{n-1} \neq 0$ and let $\alpha \in V$ be any vector such that $N^{n-1}\alpha \neq 0$. prove that α is a cyclic vector for N.

Proof Since N is nilpotent operator, $N^k = 0$ for some $k \geq 0$, therefore the minimal polynomial of N must be power of x. But, since $N^{n-1} \neq 0$, so $N, N^2, \cdots, N^{n-2}$ are not zero, by the Cayley-Hamilton theorem, $N^n = 0$.

For the given $\alpha \in V$, $N^{n-1}\alpha$ implies that $\alpha, N\alpha, N^2\alpha, \cdots, N^{n-1}\alpha$ are all not zero. (if $N^{n-k}\alpha = 0$, then $N^{k-1}N^{n-k}\alpha = 0$) Moreover, these are mutually distinct, because for $0 \leq i < j \leq n-1$, $N^i\alpha = N^j\alpha$ implies $N^{j+n-j}\alpha = N^{i+n-j}\alpha = 0$, but $i+n-j < n$, it is contradiction. Moreover, these are linearly independent, suppose that

$$C_0\alpha + C_1 N\alpha + \cdots + C_{n-1}N^{n-1}\alpha = 0, \quad C_i \in F.$$

Take $N^{n-1}, N^{n-2}, \cdots, N$ to the both side, then we obtain $C_0 = 0, C_1 = 0, \cdots, C_{n-1} = 0$. So, the set $\{\alpha, \cdots, N^{n-1}\alpha\}$ is linearly independent set of n-elements in the n-dimensional space V, so proved. ☒

7. Let V be an n-dimensional, and $T: V \to V$ is diagonalizable linear operator. Then, T has a cyclic vector if and only if T has n-distinct eigenvalues.

Proof. Suppose that T has a cyclic vector $\alpha \in V$.
Since T is diagonalizable, there are linearly independent n-vectors $\alpha_1, \cdots, \alpha_n$ such that $\alpha = \alpha_1 + \cdots + \alpha_n$ and $T\alpha_i = c_i \alpha_i$ for some $c_i \in F$.
Observe that for each $k \geq 0$,

$$T^k \alpha = T^k(\alpha_1 + \cdots + \alpha_n) = \sum_{i=1}^{n} c_i^k \alpha_i.$$

These are linearly independent if and only if

$$\begin{bmatrix} 1 & c_1 & \cdots & c_1^{n-1} \\ & \vdots & & \\ 1 & c_n & \cdots & c_n^{n-1} \end{bmatrix} x = 0$$

has only trivial solution, so the matrix is invertible. It implies $c_1, \cdots, c_n$ are distinct.

Conversely, let $\mathcal{B} = \{\alpha_1, \cdots, \alpha_n\}$ be a basis of eigenvectors for T corresponding to distinct scalar values $c_1, \cdots, c_n$. Then, $\alpha = \alpha_1 + \cdots + \alpha_n$ is a cyclic vector, because: enough to show that $\{\alpha, T\alpha, \cdots, T^{n-1}\alpha\}$ is linearly independent.
Suppose that

$$\begin{aligned} x_0\alpha + x_1 T\alpha + \cdots + x_{n-1} T^{n-1}\alpha = &(x_0 + x_1 c_1 + \cdots + x_{n-1} c_1^{n-1})\alpha_1 \\ &+ (x_0 + x_1 c_2 + \cdots + x_{n-1} c_2^{n-1})\alpha_2 \\ &\vdots \\ &+ (x_0 + x_1 c_n + \cdots + x_{n-1} c_n^{n-1})\alpha_n = 0. \end{aligned}$$

Since $\alpha_1, \cdots, \alpha_n$ are linearly independent, therefore

$$x_0 + x_1 c_1 + \cdots + x_{n-1} c_1^{n-1} = \cdots = x_0 + x_1 c_n + \cdots + x_{n-1} c_n^{n-1} = 0.$$

It equals to

$$\begin{bmatrix} 1 & c_1 & \cdots & c_1^{n-1} \\ & \vdots & & \\ 1 & c_n & \cdots & c_n^{n-1} \end{bmatrix} \begin{bmatrix} x_0 \\ \vdots \\ x_{n-1} \end{bmatrix} = 0.$$

The matrix is called Vandermonde, it is invertible when $c_1, \cdots, c_n$ are distinct. Therefore, it has only trivial solution, $x_0 = \cdots = x_{n-1} = 0$. So $\{\alpha, \cdots, T^{n-1}\alpha\}$ is linearly independent.

8. Let T be a linear operator on the finite dimensional space V. If T is cyclic and T commute with a linear operator U, then U is a polynomial in T.

Proof. We want to show that: for some polynomial $g(t) \in F[t]$,

$$U\beta = g(T)\beta \quad \text{for all } \beta \in V.$$

Let $\alpha \in V$ be the cyclic vector of T. Then, $\{\alpha, T\alpha, \cdots, T^{n-1}\alpha\}$ forms basis for V. There fore, for some scalars $c_0, \cdots, c_{n-1}$,

$$U\alpha = c_0\alpha + c_1 T\alpha + \cdots + c_{n-1}T^{n-1}\alpha.$$

Now, put $g(t) = c_0 + c_1 t + \cdots + c_{n-1}t^{n-1}$. Then

$$U\alpha = g(T)\alpha,$$

$$UT\alpha = TU\alpha = Tg(T)\alpha = g(T)T\alpha$$

$$\vdots$$

$$UT^{n-1}\alpha = g(T)T^{n-1}\alpha$$

Thus, $U = g(T)$. ▨